Eyes for the Blinds

A Pure Software-Based Assistive Navigation Framework

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Addressing the Global Challenge

2.2B

People globally with visual impairments facing mobility challenges.

91.7%

Object detection accuracy achieved by the framework.

Hardware-Minimal Solution

Independent mobility remains a significant hurdle. Existing solutions are often cost-prohibitive and bulky.

The proposed system introduces a low-cost, software-only solution utilizing a single monocular camera. It effectively captures the environment and ensures user safety by delivering robust object detection and highly precise distance metrics.

- → Distance precision maintained within $\pm5-10$ cm.
- → Operates efficiently on standard CPU-only edge devices.

The Navigation Gap

The Context

Personal autonomy heavily relies on safe and reliable navigation. Identifying surroundings and predicting obstacles is a fundamental requirement for independent mobility.

The Gap

Traditional white canes lack semantic awareness. Guide dogs are high-maintenance. Existing Electronic Travel Aids (ETAs) are often expensive and bulky due to LiDAR or ultrasonic sensor dependencies.

The Innovation

Transitioning from hardware-dependent sensors to a pure software perception loop, enabling powerful semantic navigation using accessible and ubiquitous computing devices.

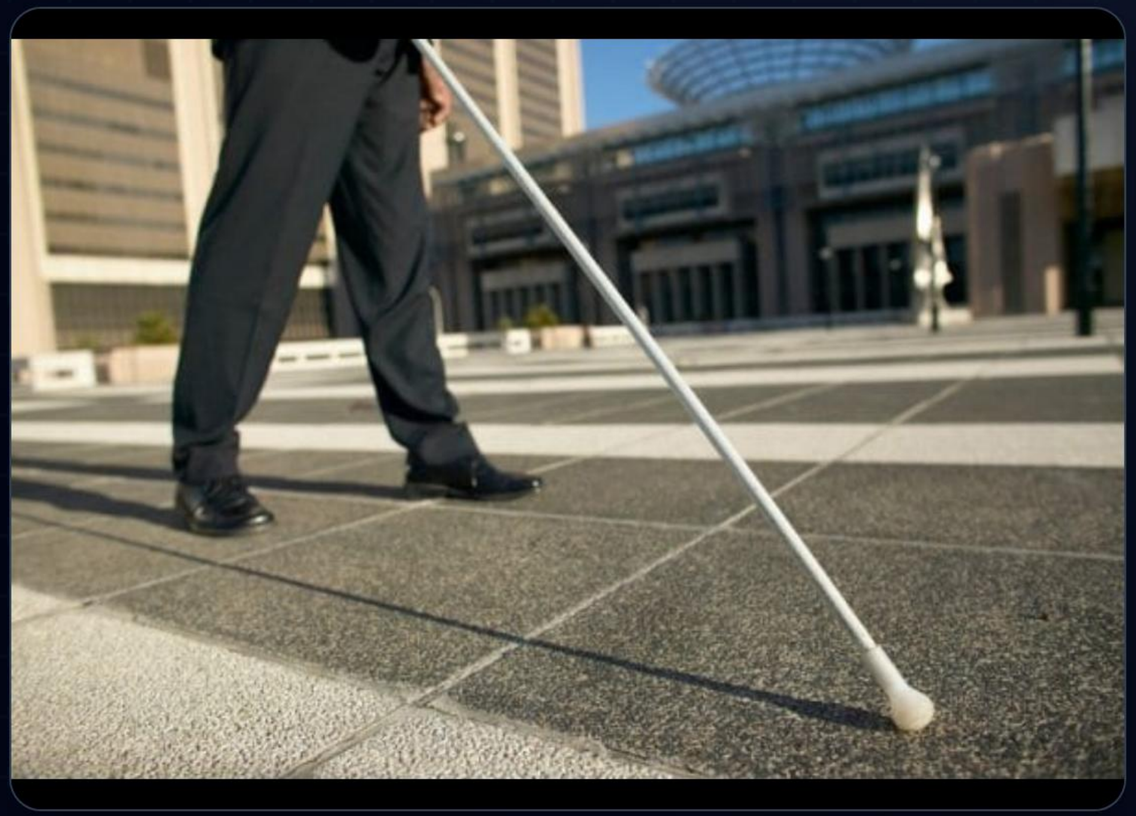
Overcoming Scale Ambiguity

The Core Problem

Scale ambiguity in monocular vision is a critical hurdle. It makes it exceptionally difficult to calculate accurate metric distances from a single 2D RGB image without stereoscopic lenses or active ranging sensors.

Framework Objectives

- → **Objective 1:** Eliminate specialized ranging hardware (like LiDAR or depth sensors) to drastically reduce system cost and physical weight.
- → **Objective 2:** Implement real-time, robust semantic depth awareness capable of running smoothly on standard CPU-only devices.



Software-Centric Pipeline Architecture

2. Parallel Processing

The system bifurcates the video feed, running object detection and depth estimation networks simultaneously for optimal speed.

1. Video Input Stream

Captures a continuous 640×488 RGB video stream directly from a monocular camera, functioning at a consistent 12–18 FPS.

3. Edge Optimization

Both perception models are heavily optimized using ONNX-based quantization to ensure high-performance execution on CPU-only edge environments.

Dual AI Model Integration

Semantic Detection

Powered by YOLOv8n (Nano)

The system utilizes an anchor-free design which vastly improves the detection capabilities for irregular, thin objects like street poles and signposts.

- → Recognizes over 80+ object classes.
- → Optimized for pedestrians, vehicles, and standard urban obstacles.
- → Extremely low computational footprint.

Depth Estimation

Powered by MiDaS_small

Provides dense, pixel-wise relative depth mapping across the entire visual frame, acting as a critical safety net.

- → Catches unclassified or novel environmental hazards.
- → Translates 2D images into topological maps.
- → Crucial for general scene understanding outside YOLO's vocabulary.

The Core Innovation: Hybrid Estimation

Bridging the Gap

The framework merges two distinct methodologies to overcome monocular scale ambiguity, ensuring precision without physical sensors.

- → **Geometric Lens Modeling:** Used for known objects detected by YOLOv8n. It leverages the camera's intrinsic parameters and the known real-world heights of objects.
- → **Inverse Depth Mapping:** Applied to general scenes and unclassified objects mapped by MiDaS. It uses calibrated scaling constants to convert relative depth arrays into absolute metric distances.

This fusion drastically reduces error margins in critical navigation zones.

Geometric Lens Formula

$$D = \frac{f \times H_{real} \times H_{image_res}}{H_{pixel} \times H_{sensor_phys}}$$

Inverse Mapping Formula

$$D = \frac{k}{DepthValue + \varepsilon}$$

Intuitive Routing & Audio Feedback

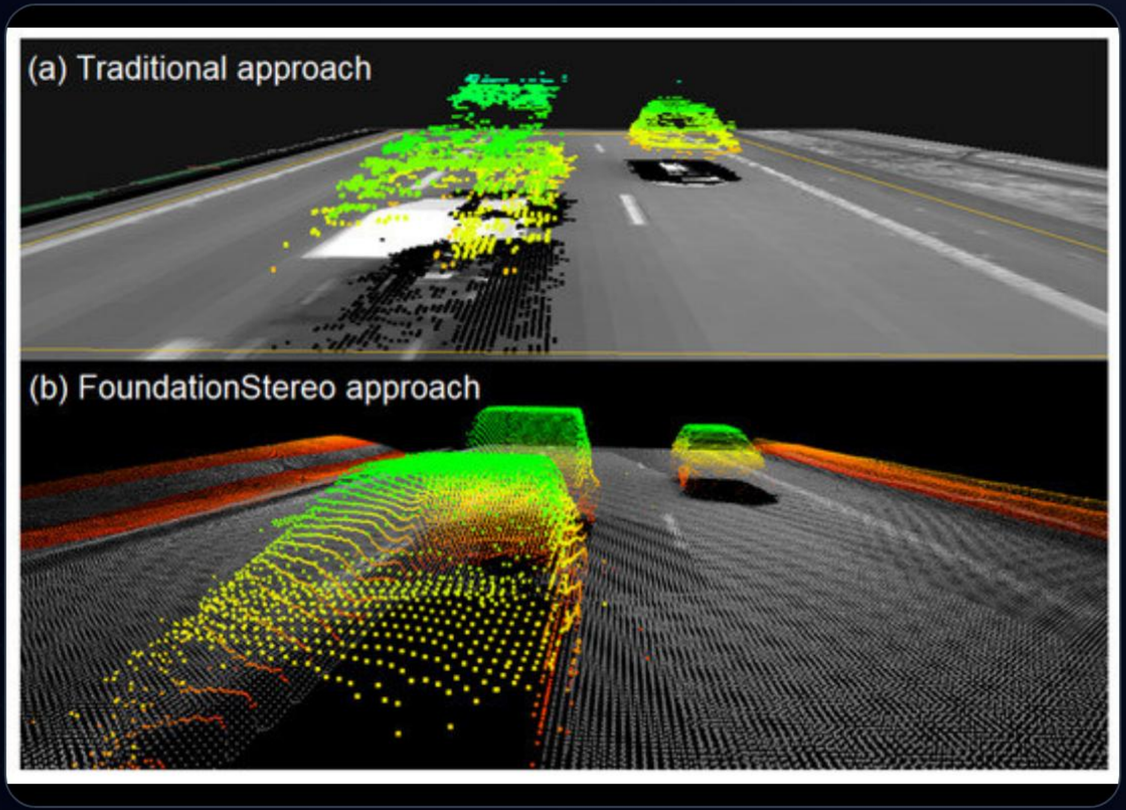
Spatial Processing

The visual field is dynamically segmented into Left, Center, and Right zones. If the central path is blocked by an obstacle closer than 1.5m, the logic engine computes and suggests the path with maximum clearance.

Human-Centric Audio

Metric distances are difficult to gauge without vision. The system translates distances into "footsteps" (standardized at 0.75m per step) to drastically reduce cognitive load.

- → **Tiered Alerts:** Urgency dictates the audio tone and priority.
- → *Example:* "STOP! Car less than one step away."



Performance Evaluation

Accuracy & Efficiency

The system was rigorously tested in real-world environments, proving the viability of a pure software approach.

- **Distance Accuracy:** Errors successfully constrained within just $\pm5-10$ cm inside the highly critical 0–2 m safety zone.
- **Processing Speed:** Maintains a consistent 12–18 FPS purely on CPU hardware.
- **Latency:** End-to-end response time is kept strictly under 500ms.

Hardware Comparison

Benchmarked against traditional physical sensor arrays, the hybrid software model presents a disruptive advantage.

- Performance is highly comparable to heavy hardware-assisted systems like LiDAR.
- Eliminates sensor dead-zones found in ultrasonic arrays.
- Achieved at a small fraction of the manufacturing cost and physical weight.

The Path Forward

Conclusion

We successfully validated a highly scalable, software-only framework that drastically lowers the barrier to entry for independent mobility aids.

Future Work

The next iteration aims to integrate Large Vision-Language Models (LVLMs) to provide rich, contextual understanding of the environment beyond simple obstacle avoidance (e.g., "Empty seat on your left").

Ultimate Goal

Deployment of this software stack onto wearable mobile platforms and smart glasses, delivering real-world, hands-free accessibility for the visually impaired.

Image Sources

https://brailleworks.com/wp-content/uploads/2018/10/Stock-image-White-Cane-1.jpg

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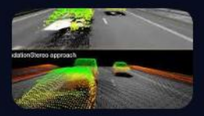

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